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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING H01																A	
																B	
																C	
																D	
																E	
F																F	
				Date Prep.	15.11.2018	PROJECT TITLE				ABB EPMV		COVER SHEET		Scale	NTS	= H01	
				Prepared	YR	MV Switchgear for MNA Serang Plant						MVD-24 20kV Compressor		Customer	PT Multimas Nabati Asahan	& 304	
00	Issued for Approval	15.01.19	YR	Checked	AW					PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-304.06	Sh. 1	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042						SBC-W INCOMING		Customer Doc. Nbr.		Cont. 28	
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	==MVD-24=H01&304/2			Index of Sheet				00							
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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING		Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant		Customer					PT Multimas Nabati Asahan	& 304	
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.			18.042-MNA-304.06	Sh. 2	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042					Customer Doc. Nbr.		Cont. 28		
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REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS

(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)

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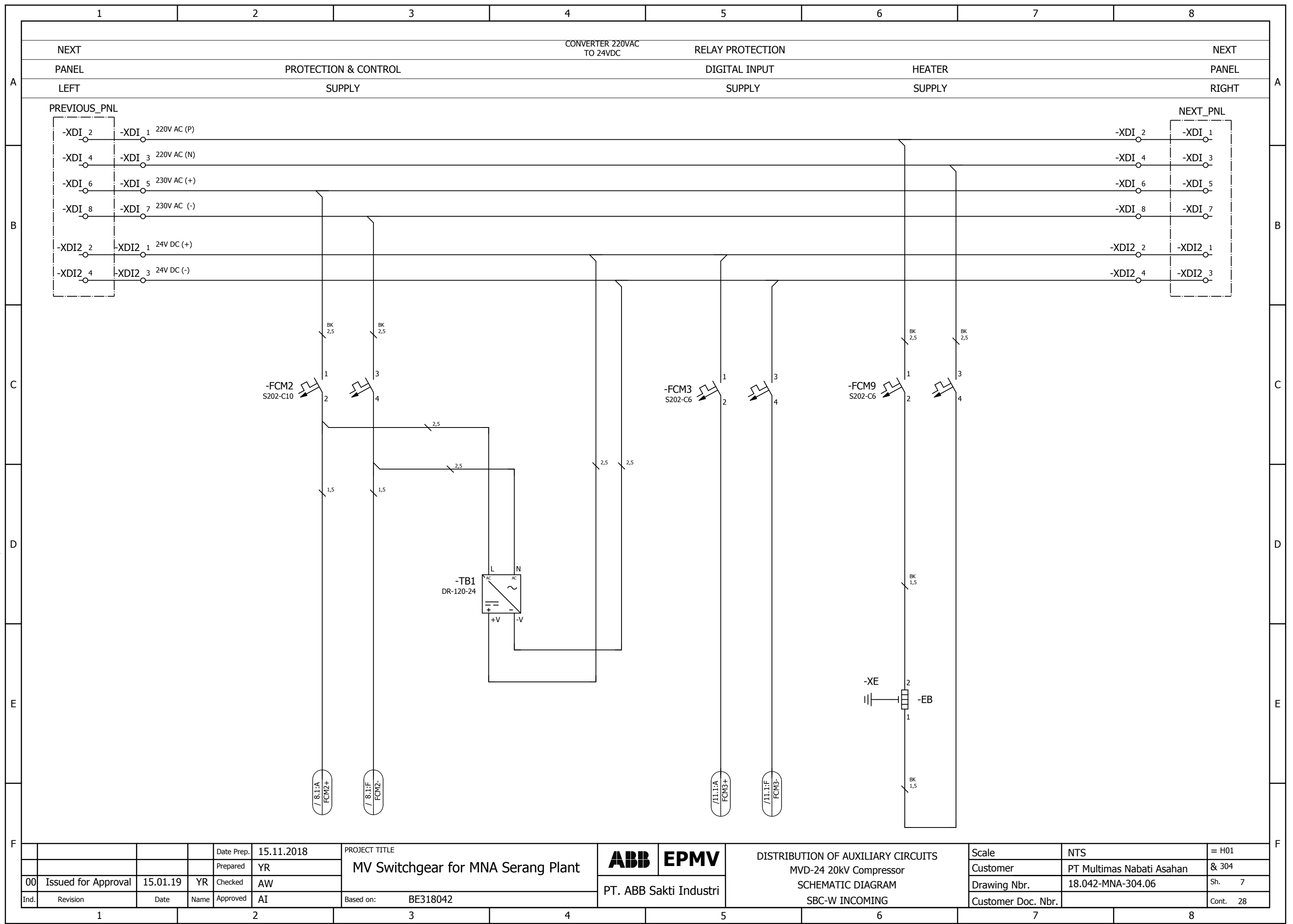
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
B	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							B
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
C	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							C
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
D	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							D
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
E	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							E
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
F	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							F
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
G	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							G
	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
H	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							H
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
I	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							I
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
J	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							J
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
K	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							K
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
L	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							L
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
M								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							M
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
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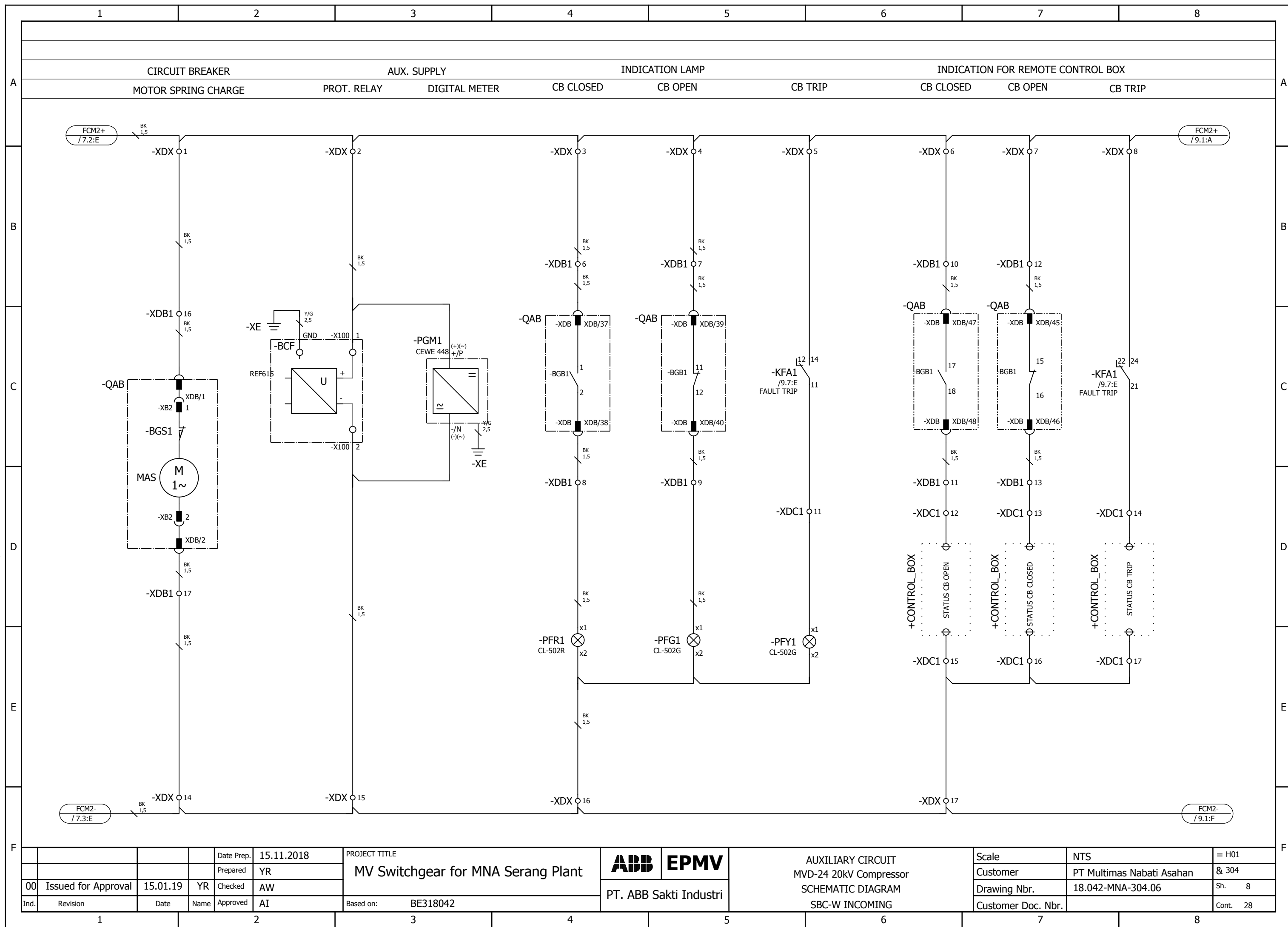
A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS							
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS							
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS							
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK							
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING							
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH							
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES							
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES							
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS							
	-PFF	FLAG RELAYS						-RAR	RESISTORS							
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS							
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE							
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION							
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD							
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2							
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2							
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS							
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4							
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5							
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES							
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS							
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS							
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS							
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS							
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES							
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS							
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS							
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES							
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS							
	-PGV	VOLTMETERS						-TB	CONVERTER							
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE							
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE							
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE							
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE							
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER							
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														
F	-QBD4	TWO-WAY DISCONNECTOR														
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER														
					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-24 20kV Compressor		Scale	NTS		= H01
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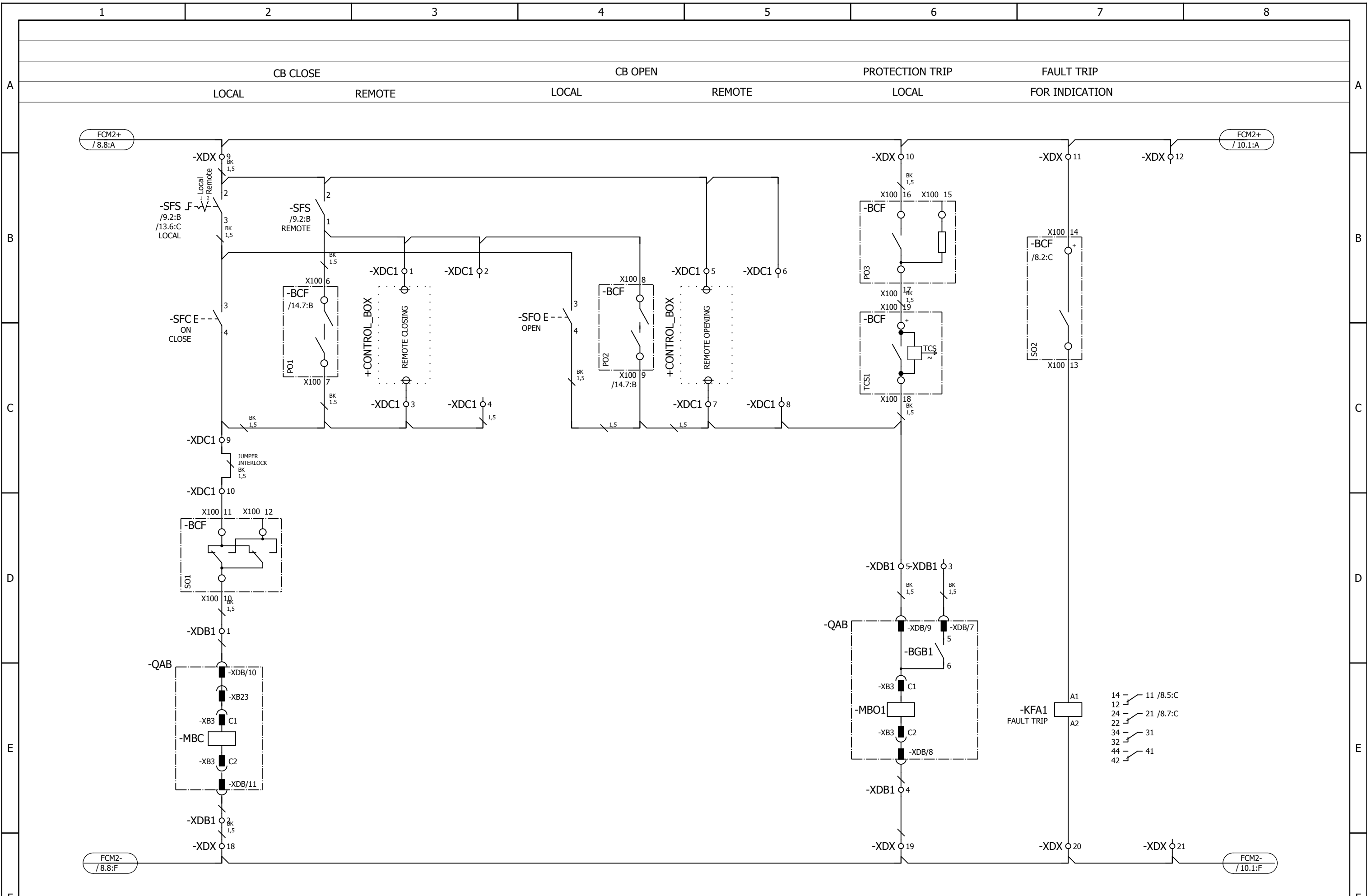
A	-TFA	ACTIVE POWER TRANSDUCERS	-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A										
	-TFC	CURRENT TRANSDUCERS	-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING											
	-TFF	FREQUENCY TRANSDUCERS	-XDI3	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION											
	-TFJ	ACTIVE ENERGY TRANSDUCERS	-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS											
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XDI6	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS											
	-TFM	MULTIFUNCTION TRANSDUCERS													
	-TFP	POWER-FACTOR TRANSDUCERS													
	-TFQ	REACTIVE POWER TRANSDUCERS													
	-TFS	SIGNAL CONVERTERS													
	-TFV	VOLTAGE TRANSDUCERS													
B	-WA	BUSBARS	-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS	B										
	-WBC	POWER CABLES	-XDI8	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS											
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDI9	TERMINAL BLOCK FOR SPECIAL USE											
	-WGC	CONTROL CABLES	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD											
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS											
	-WH	OPTICAL FIBRES	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT											
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDS	CURRENT SOCKETS											
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB											
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK											
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK											
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS	C										
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1											
			-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2											
			-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3											
			-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR											
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XDX	SUPPORT TERMINAL BLOCK											
			-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE											
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE											
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE											
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE											
D	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE	D										
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE											
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT											
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET													
	-XDC	CUSTOMER TERMINAL BLOCK													
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER													
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR													
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT													
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS													
E	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			E										
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE													
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE													
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2													
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2													
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS													
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR													
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS													
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS													
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)													
F	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)			F										
1		2		3		4		5		6		7		8	
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING		Scale	NTS		= H01
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 304
00	Issued for Approval	15.01.19	YR	Checked	AW							Drawing Nbr.	18.042-MNA-304.06		Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri				Customer Doc. Nbr.			Cont. 28



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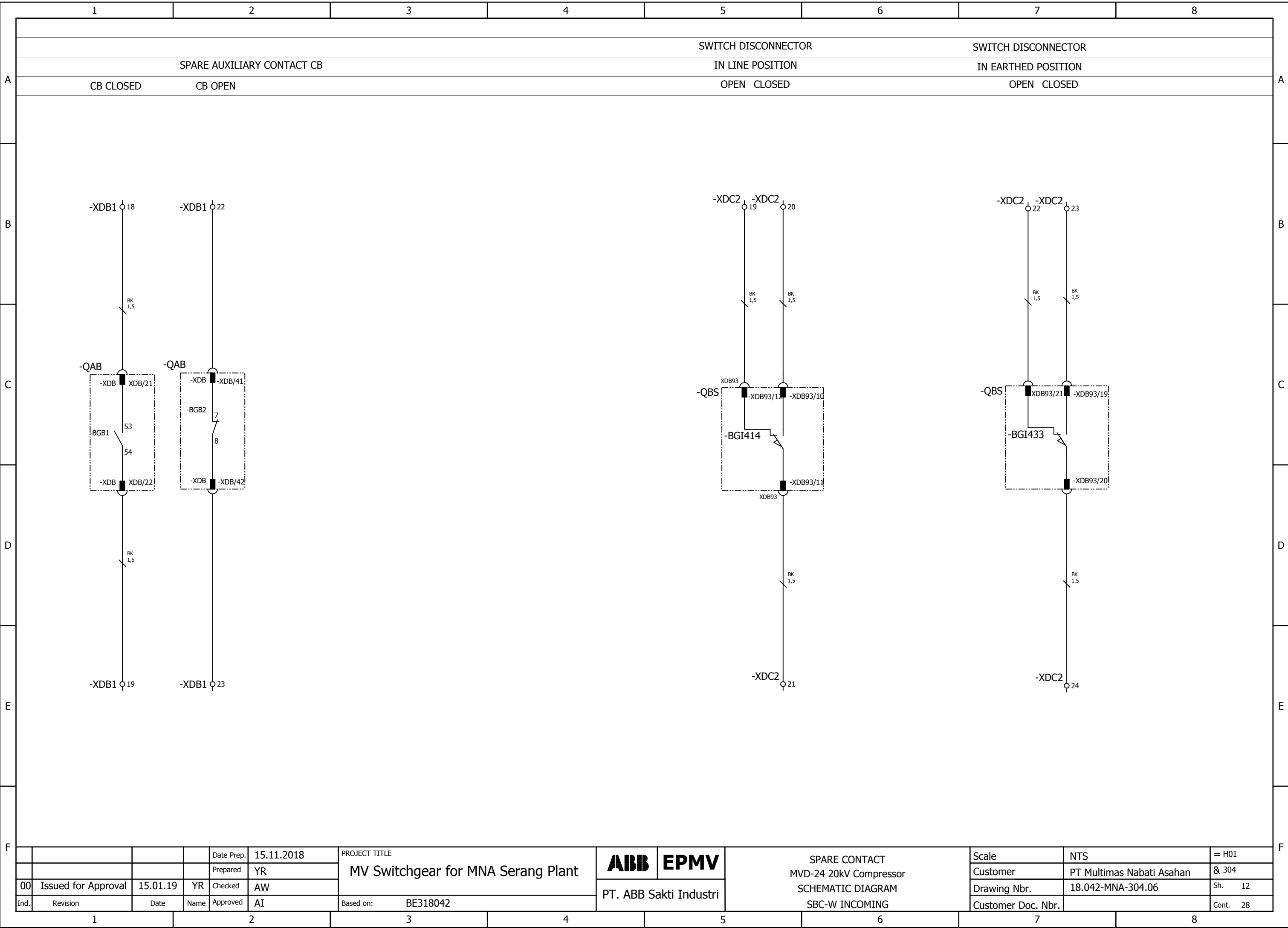


		Date Prep.		15.11.2018	PROJECT TITLE		ABB EPMV		CIRCUIT BREAKER CONTROL MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING		Scale	NTS	= H01
		Prepared		YR							Customer	PT Multimas Nabati Asahan	& 304
00 Issued for Approval		15.01.19	YR	Checked							Drawing Nbr.	18.042-MNA-304.06	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 28

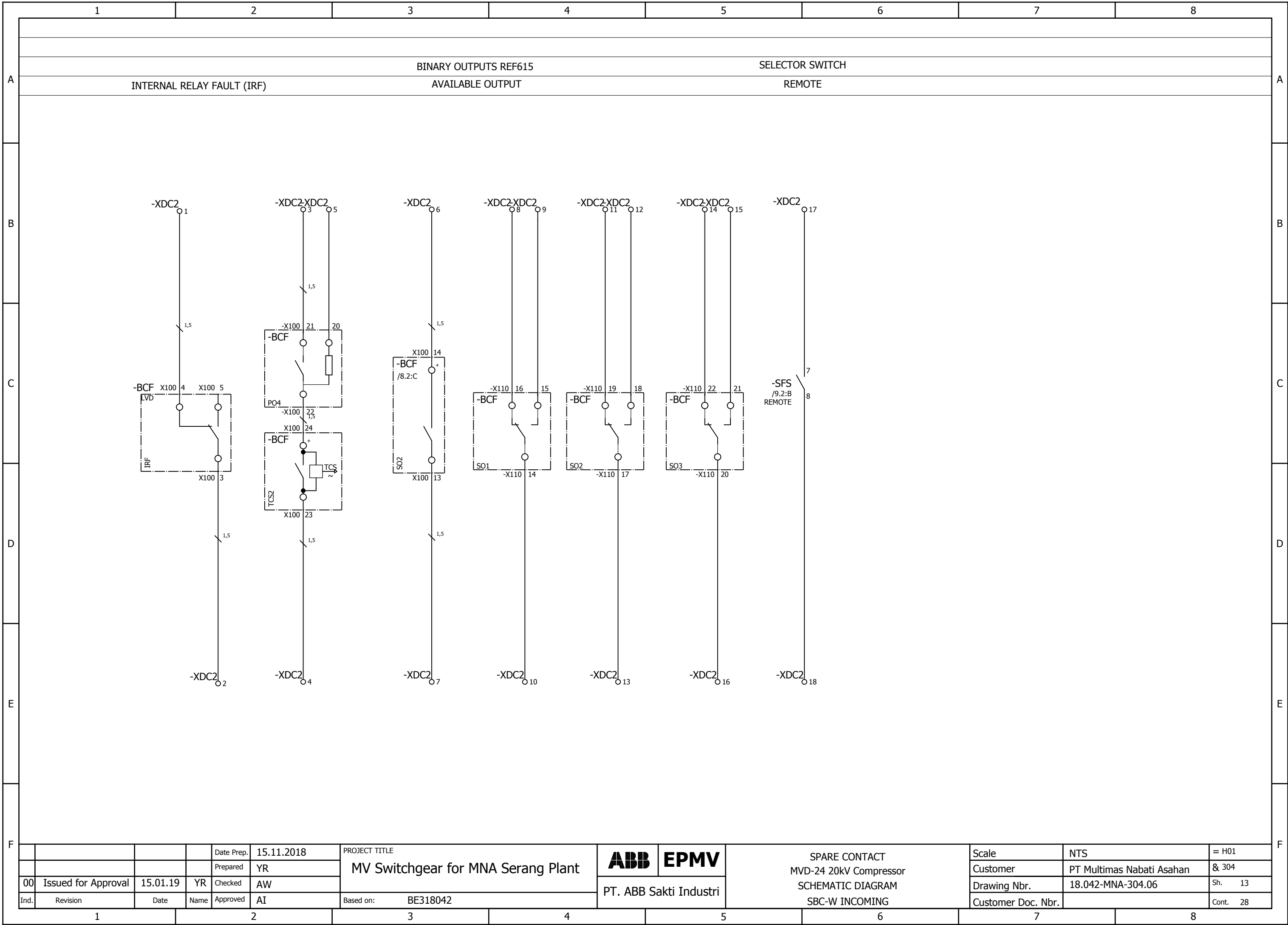
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-304.06		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.			Cont. 28	
							PT. ABB Sakti Industri					

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-304.06	Sh. 11	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 28	

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00	Issued for Approval	15.01.19	YR	Date Prep.	15.11.2018	PROJECT TITLE	MV Switchgear for MNA Serang Plant	Based on:	BE318042	ABB	EPMV	PT. ABB Sakti Industri	SPARE CONTACT MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR									Customer	PT Multimas Nabati Asahan	& 304
				Checked	AW									Drawing Nbr.	18.042-MNA-304.06	Sh. 12
				Approved	AI									Customer Doc. Nbr.		Cont. 28



This is a detailed schematic diagram of a 20kV MV Switchgear for a Compressor Motor (MVA) at the MNA Serang Plant. The diagram illustrates the electrical connections between the incoming 20kV bus (L1, L2, L3), the switchgear components (BCT1, BCT2, BCT3, XDA, XTA1, XTA2), and the motor (M). It also shows the internal wiring for the motor protection system (MPS) and the motor control system (MCS).

Legend:

- Terminal IS CLOSED
- Terminal IS OPEN

Key Components and Connections:

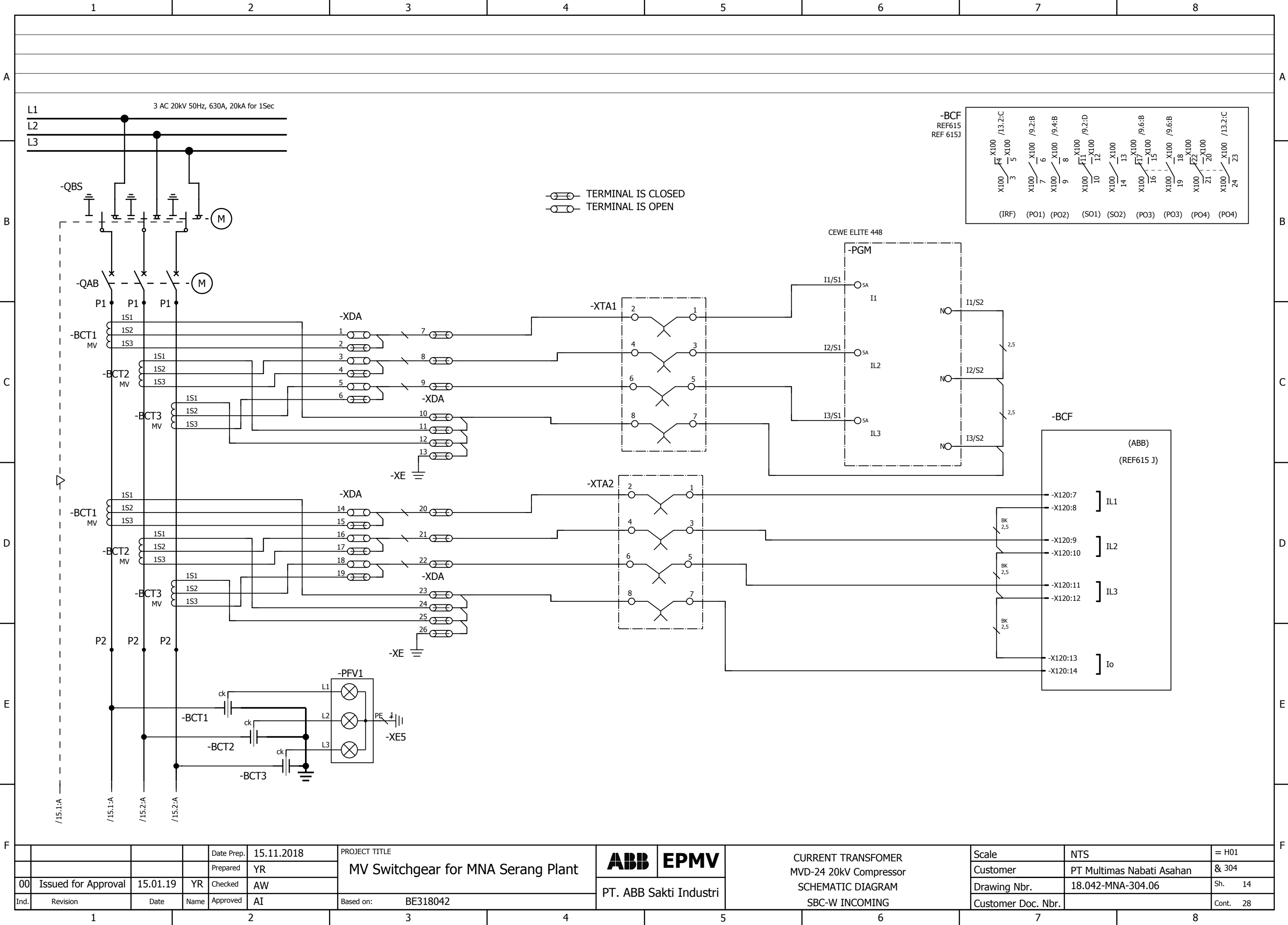
- Busbar:** 3 AC 20kV 50Hz, 630A, 20kA for 1Sec.
- Switchgear:** BCT1, BCT2, BCT3 (MV), XDA, XTA1, XTA2.
- Motor:** M (Compressor Motor).
- Motor Protection System (MPS):** -QBS, -QAB, -BCT1, -BCT2, -BCT3, -PFV1, -XE, -XE5.
- Motor Control System (MCS):** -PGM, -BCF, -BCF (ABB REF615 J), -X120:7, -X120:8, -X120:9, -X120:10, -X120:11, -X120:12, -X120:13, -X120:14.

Terminal Block Connections:

- Terminal 1:** L1, L2, L3
- Terminal 2:** P1, P2
- Terminal 3:** P1, P2
- Terminal 4:** P1, P2
- Terminal 5:** P1, P2
- Terminal 6:** P1, P2
- Terminal 7:** P1, P2
- Terminal 8:** P1, P2
- Terminal 9:** P1, P2
- Terminal 10:** P1, P2
- Terminal 11:** P1, P2
- Terminal 12:** P1, P2
- Terminal 13:** P1, P2
- Terminal 14:** P1, P2
- Terminal 15:** P1, P2
- Terminal 16:** P1, P2
- Terminal 17:** P1, P2
- Terminal 18:** P1, P2
- Terminal 19:** P1, P2
- Terminal 20:** P1, P2
- Terminal 21:** P1, P2
- Terminal 22:** P1, P2
- Terminal 23:** P1, P2
- Terminal 24:** P1, P2
- Terminal 25:** P1, P2
- Terminal 26:** P1, P2

Notes:

- The diagram is a schematic representation of the switchgear and motor control system.
- The motor (M) is a 20kV motor.
- The motor protection system (MPS) is designed to protect the motor from overcurrent, short circuit, and other faults.
- The motor control system (MCS) is designed to start and stop the motor.

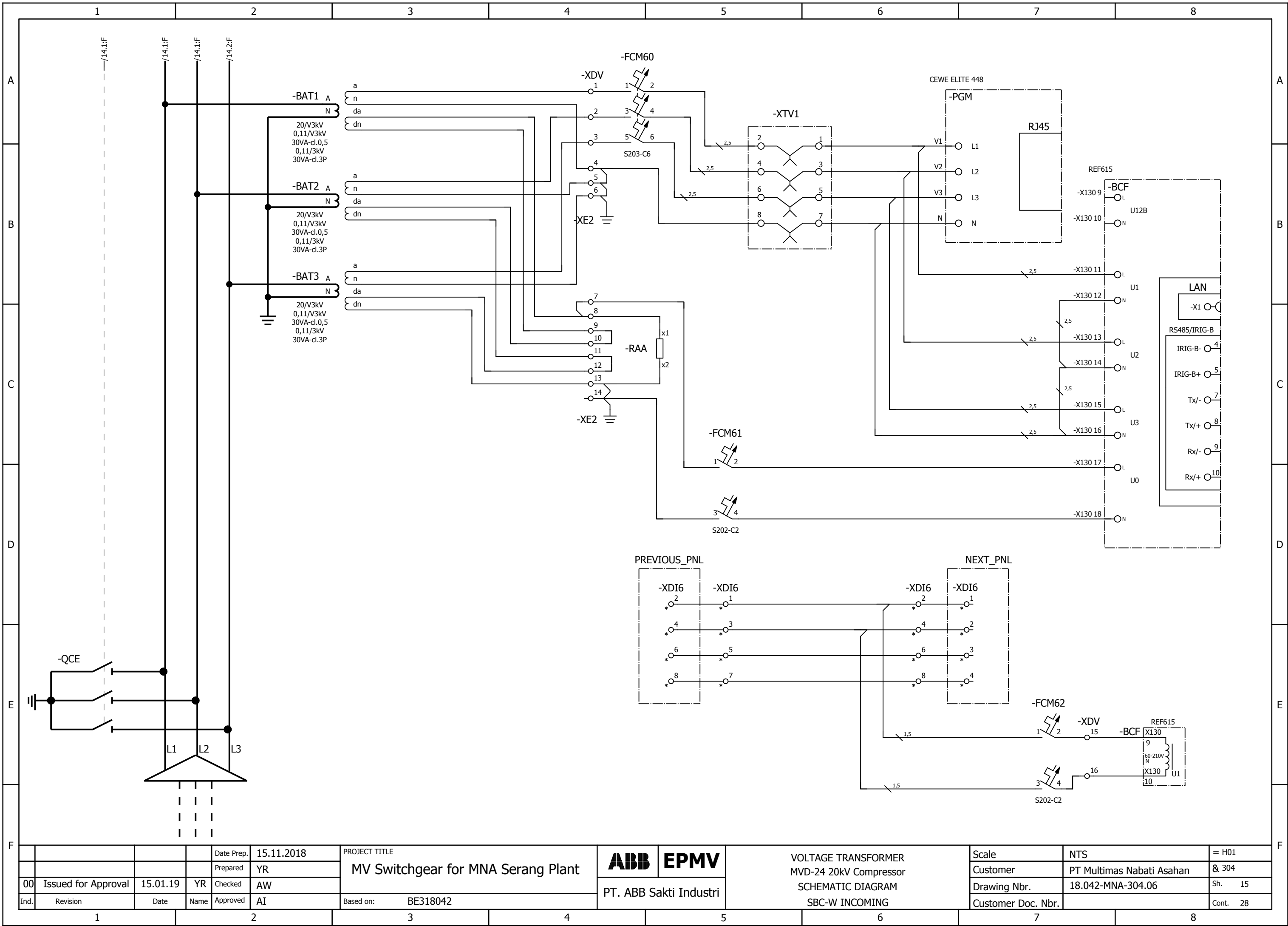


This is a detailed schematic diagram of a 20kV MV Switchgear for a Compressor, specifically for the MNA Serang Plant. The diagram illustrates the electrical connections between the incoming 3 AC 20kV 50Hz supply (630A, 20kA for 1Sec) and the internal components of the switchgear.

Key Components and Connections:

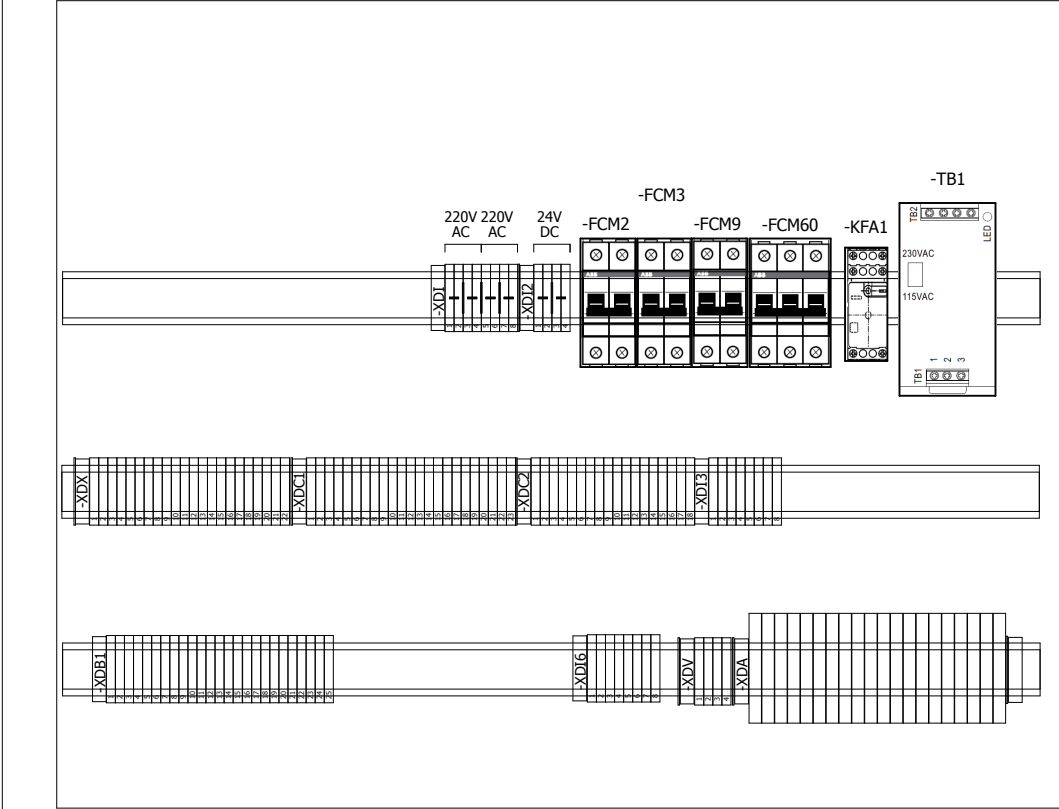
- Supply and Protection:** The supply enters through busbars L1, L2, and L3. It is protected by a circuit breaker (-QBS) and a main switch (-QAB). The switchgear is divided into three main sections (P1, P2, P3) with associated busbars and circuit breakers (-BCT1, -BCT2, -BCT3).
- Interlocks and Status:** The diagram includes interlocking mechanisms (-XDA, -XTA1, -XTA2) and status indicators (-XE, -XE5) to ensure safe operation.
- Current Transformers (CTs):** CTs are connected to the busbars and the main switch to monitor current levels.
- Control and Protection:** The control system (-PGM) is connected to the switchgear through a CEWE ELITE 448 interface. It includes status indicators (I1, I2, I3) and control lines (I1/S1, I2/S1, I3/S1).
- Terminal Block and Connections:** The terminal block (-BCF) is connected to the switchgear through a series of interlocking mechanisms (-XDA, -XTA1, -XTA2) and status indicators (-XE, -XE5). The terminal block is labeled with various connection points (X100, X101, X102, X103, X104, X105, X106, X107, X108, X109, X110, X111, X112, X113, X114, X115, X116, X117, X118, X119, X120, X121, X122, X123, X124, X125, X126, X127, X128, X129, X130, X131, X132, X133, X134, X135, X136, X137, X138, X139, X140, X141, X142, X143, X144, X145, X146, X147, X148, X149, X150, X151, X152, X153, X154, X155, X156, X157, X158, X159, X160, X161, X162, X163, X164, X165, X166, X167, X168, X169, X170, X171, X172, X173, X174, X175, X176, X177, X178, X179, X180, X181, X182, X183, X184, X185, X186, X187, X188, X189, X190, X191, X192, X193, X194, X195, X196, X197, X198, X199, X200, X201, X202, X203, X204, X205, X206, X207, X208, X209, X210, X211, X212, X213, X214, X215, X216, X217, X218, X219, X220, X221, X222, X223, X224, X225, X226, X227, X228, X229, X230, X231, X232, X233, X234, X235, X236, X237, X238, X239, X240, X241, X242, X243, X244, X245, X246, X247, X248, X249, X250, X251, X252, X253, X254, X255, X256, X257, X258, X259, X260, X261, X262, X263, X264, X265, X266, X267, X268, X269, X270, X271, X272, X273, X274, X275, X276, X277, X278, X279, X280, X281, X282, X283, X284, X285, X286, X287, X288, X289, X290, X291, X292, X293, X294, X295, X296, X297, X298, X299, X300, X301, X302, X303, X304, X305, X306, X307, X308, X309, X310, X311, X312, X313, X314, X315, X316, X317, X318, X319, X320, X321, X322, X323, X324, X325, X326, X327, X328, X329, X330, X331, X332, X333, X334, X335, X336, X337, X338, X339, X340, X341, X342, X343, X344, X345, X346, X347, X348, X349, X350, X351, X352, X353, X354, X355, X356, X357, X358, X359, X360, X361, X362, X363, X364, X365, X366, X367, X368, X369, X370, X371, X372, X373, X374, X375, X376, X377, X378, X379, X380, X381, X382, X383, X384, X385, X386, X387, X388, X389, X390, X391, X392, X393, X394, X395, X396, X397, X398, X399, X400, X401, X402, X403, X404, X405, X406, X407, X408, X409, X410, X411, X412, X413, X414, X415, X416, X417, X418, X419, X420, X421, X422, X423, X424, X425, X426, X427, X428, X429, X430, X431, X432, X433, X434, X435, X436, X437, X438, X439, X440, X441, X442, X443, X444, X445, X446, X447, X448, X449, X450, X451, X452, X453, X454, X455, X456, X457, X458, X459, X460, X461, X462, X463, X464, X465, X466, X467, X468, X469, X470, X471, X472, X473, X474, X475, X476, X477, X478, X479, X480, X481, X482, X483, X484, X485, X486, X487, X488, X489, X490, X491, X492, X493, X494, X495, X496, X497, X498, X499, X500, X501, X502, X503, X504, X505, X506, X507, X508, X509, X510, X511, X512, X513, X514, X515, X516, X517, X518, X519, X520, X521, X522, X523, X524, X525, X526, X527, X528, X529, X530, X531, X532, X533, X534, X535, X536, X537, X538, X539, X540, X541, X542, X543, X544, X545, X546, X547, X548, X549, X550, X551, X552, X553, X554, X555, X556, X557, X558, X559, X560, X561, X562, X563, X564, X565, X566, X567, X568, X569, X570, X571, X572, X573, X574, X575, X576, X577, X578, X579, X580, X581, X582, X583, X584, X585, X586, X587, X588, X589, X590, X591, X592, X593, X594, X595, X596, X597, X598, X599, X600, X601, X602, X603, X604, X605, X606, X607, X608, X609, X610, X611, X612, X613, X614, X615, X616, X617, X618, X619, X620, X621, X622, X623, X624, X625, X626, X627, X628, X629, X630, X631, X632, X633, X634, X635, X636, X637, X638, X639, X640, X641, X642, X643, X644, X645, X646, X647, X648, X649, X650, X651, X652, X653, X654, X655, X656, X657, X658, X659, X660, X661, X662, X663, X664, X665, X666, X667, X668, X669, X670, X671, X672, X673, X674, X675, X676, X677, X678, X679, X680, X681, X682, X683, X684, X685, X686, X687, X688, X689, X690, X691, X692, X693, X694, X695, X696, X697, X698, X699, X700, X701, X702, X703, X704, X705, X706, X707, X708, X709, X710, X711, X712, X713, X714, X715, X716, X717, X718, X719, X720, X721, X722, X723, X724, X725, X726, X727, X728, X729, X730, X731, X732, X733, X734, X735, X736, X737, X738, X739, X740, X741, X742, X743, X744, X745, X746, X747, X748, X749, X750, X751, X752, X753, X754, X755, X756, X757, X758, X759, X760, X761, X762, X763, X764, X765, X766, X767, X768, X769, X770, X771, X772, X773, X774, X775, X776, X777, X778, X779, X780, X781, X782, X783, X784, X785, X786, X787, X788, X789, X790, X791, X792, X793, X794, X795, X796, X797, X798, X799, X800, X801, X802, X803, X804, X805, X806, X807, X808, X809, X810, X811, X812, X813, X814, X815, X816, X817, X818, X819, X820, X821, X822, X823, X824, X825, X826, X827, X828, X829, X830, X831, X832, X833, X834

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	VOLTAGE TRANSFORMER MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01	
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 304	
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri			Drawing Nbr.	18.042-MNA-304.06	Sh. 15	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.					Cont. 28	

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK
10	-XTV1 (TEST BLOCK , YONGSUNG)	-XTV, VOLTAGE TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR							Customer
00	Issued for Approval	15.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-304.06	Sh. 17
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 28

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	1	2	3	4	5	6	7	8
A	Summarized parts list							IDABB_F02_005-IDABB
	Carry over: 0.00							
	TAG ID	DESCRIPTION			MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS
	-BCF	Feeder Management Relay HBFJAEAGABC1ANA11G			ABB	REF615J HBFJAEAGABC1ANA11G	2	/8.2:C;/15.7:B
	-EB	HEATER 230VAC, 50W			SANYANG	CDXN010033P0009	1	/7.6:E
	-FCM2	MCB S202-C10 MCB. 2P, 10A			ABB	2CDS252001R0104	1	/7.2:C
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A			ABB	2CDS252001R0064	2	/7.5:C;/7.6:C
	-FCM60	MCB S203-C6 MCB. 3P, 6A			ABB	2CDS253001R0064	1	/15.4:A
	-FCM61;-FCM62	MCB S202-C2 MCB. 2P, 2A			ABB	2CDS252001R0024	2	/15.5:D;/15.7:E
	-KFA1	Auxiliary Relay 230AC 4COS			PHOENIX	1YRA3DL266230001	1	/9.7:E
	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G			ABB	1SFA619402R5232	1	/8.5:E
	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R			ABB	1SFA619402R5231	1	/8.4:E
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y			ABB	1SFA619402R5233	1	/8.6:E
	-PGM1	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC			CEWE	1YRA4EA0350001	1	/8.3:C
	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR			IDEC	1YRA3DA14700001	1	/9.2:B
	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR			IDEC	1YRA3DA14700002	1	/9.4:B
	B	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)			K&N	1YRA3DA17940011	1
-TB1		Voltage Input : 85-264 VAC/120-370V DC Voltage Output : 24 VDC, 5A			MEANWELL	DR-120-24	1	/7.3:D
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSCTT-04C	2	/14.4:B;/14.4:D
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting			YONGSUNG	YSPTT-04C	1	/15.5:A
C								
D								
E								
F								

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST MVD-24 20kV Compressor SCHEMATIC DIAGRAM SBC-W INCOMING	Scale	NTS	= H01
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 304
00	Issued for Approval	15.01.19	YR	Checked	AW			Drawing Nbr.	18.042-MNA-304.06		Sh. 18		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 28		

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[illegible]

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F		E		D		C		B		A	

[illegible]

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A		B		C		D		E		F		A		B	
F		F		F		F		F		F		F		F	
Ind.		Revision		Date		Name		Approved		AI		Based on:		BE318042	
				Date Prep.		15.11.2018		PROJECT TITLE		MV Switchgear for MNA Serang Plant		ABB EPMV		Terminal diagram ==MVD-24=H01+-XDI2	
				Prepared		YR								Scale	
				Checked		AW								Customer	
				Approved										Drawing Nbr.	
														Customer Doc. Nbr.	
														= H01	
														& 304	
														Sh. 24	
														Cont. 28	
1		2		3		4		5		6		7		8	

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